7.4.4 Neighbors

Because we know when features share arcs, we also know what features neighbor each other. The function `Topojson.neighbors` builds an array of all the features that share a border. We can use this array to easily identify neighboring countries in our dataset using the code in the following listing. The results of the interaction provided by this code are shown in figure 7.19.

Listing 7.18 Calculating neighbors and interactive highlighting

```
    var neighbors =
topojson.neighbors(topoCountries.objects.countries.geometries);

    d3.selectAll("path.countries")
    .on("mouseover", findNeighbors)
    .on("mouseout", clearNeighbors);

    function findNeighbors (d,i) {
d3.select(this).style("fill", "red");
d3.selectAll("path.countries")
    .filter(function (p,q) {return neighbors[i].indexOf(q) > -1})
 .style("fill", "green")
    };

    function clearNeighbors () {
d3.selectAll("path.countries").style("fill", "gray");
    };
```

Creates an array indicating neighbors by their array position

Colors the country you hover over red

Colors all neighbors green

Colors all countries gray to "clear" results

TopoJSON is a powerful new technology that provides tremendous opportunity for web map development. Understanding how it models data and the functionality that it provides are key to creating maps that impress users. As you explore traditional web



Figure 7.19 Hover behavior displaying the neighbors of France using TopoJSON's `neighbor` function. Because Guyana is an overseas department of France, France is considered to be neighbors with Brazil and Suriname. This is because France is represented as a multipolygon in the data, and any neighbors with any of its shapes are returned as neighbors.